

Designing a Basic Network Structure

Estimated time: 10 minutes

Objectives

After completing this reading, you will be able to:

- List the benefits of network design
- Describe the consequences of skipping network design planning
- List popular network design tools
- Provide an overview of the steps used to design a basic network structure

Introduction

Network design is the process of planning a network infrastructure tailored to meet an organization's specific needs. Good network design requires selecting the correct hardware, software, protocols, and network architecture to deliver data efficiently and securely across the organization.

The process of designing a network also includes evaluating current and future network requirements, selecting network devices, and defining the overall network layout.

Let's begin by exploring the benefits of network design.

Benefits of network design

A well-thought-out network design provides several key benefits:

- **Are scalable:** Thoughtfully designed networks adapt to increased data loads and new devices without significant overhauls.
- **Are secure:** Good network design includes security measures, such as firewalls, encryption, and access controls.
- **Provide long-term cost savings:** Implementing network resources that are sized correctly for network loads enables organizations to use their resources for more time without having to continually upgrade network hardware. A well-designed network also minimizes costly downtime because of overloaded hardware.
- **Are efficient:** A well-structured network delivers data and applications quickly so that that employees can quickly access applications and data.

So, let's examine what can happen if network design planning is overlooked or skipped.

Consequences of skipping network design planning

Skipping network planning can lead to the following significant problems:

- Slow performance and frequent network downtime
- Vulnerability to cybersecurity threats
- Overspending because of the initial purchase of unnecessary or incompatible equipment.

Next, check out some popular network design tools.

Popular network design tools

Professionals use a variety of tools to optimize network design. Here are some widely used network design tools, including IBM solutions and other vendors:

- **IBM Netcool Network Management** offers comprehensive network management, including real-time monitoring, fault detection, and performance analysis.
- **IBM Cloud Pak for Network Automation** automates network deployment and management and tool integrates AI to enhance network decision-making and efficiency.
- **IBM Spectrum Control** primarily used for storage management, and provides valuable insights into network performance and resource allocation, to optimize the network's data flow and storage.
- **IBM SevOne Network Performance Management** focuses on performance monitoring and analytics, which is essential during the design phase to ensure that the network meets performance benchmarks and can handle projected loads.

And, consider some other vendor choices:

- **Cisco Network Assistant** is a tool for configuring, monitoring, and troubleshooting networks.
- **SolarWinds Network Topology Mapper** automatically maps out network devices and connections.
- **NetBrain** provides network automation and real-time mapping capabilities.
- **Microsoft Visio** creates detailed network diagrams.
- **GNS3** This is an open-source tool for simulating complex networks,
- **Wireshark** is a network protocol analyzer for monitoring network traffic in real time.

Now that you know why network design is important, explore the steps needed to design a basic network structure.

Steps to design a basic network structure

- **Gather the business requirements.**
 - Identify why the organization needs to install or update a network.
 - Define the budget and infrastructure constraints
 - Determine which legacy systems need support.
- **Assess the state of the current network hardware and technologies.**
 - Evaluate the existing network infrastructure.
 - Diagram and document current network devices, cabling layouts, and communication technologies.

- Decide which network hardware the organization can reuse and which hardware the organization must replace.
- **Define the new network's goals.**
 - Document features for the new network, such as improved performance, enhanced security, or simplified management.
- **Select the devices and technologies.**
 - Specify the necessary hardware and software, including servers, routers, firewalls, network management and security software
- **Design the physical network.**
 - Plan for the servers, switches, firewalls, cabling, and more.
 - Use diagrams to map the hardware connections for network structure compatibility
 - Develop an implementation plan that specifies timelines and the resources (people) needed for implementation.
- **Document the network design.**
 - Store the network documentation, including the diagrams, hardware configuration information, and business requirements, in an accessible location. The company will need this documentation for future modifications, troubleshooting, and disaster recovery.

By leveraging these tools, including IBM's suite of network management and automation solutions, organizations can ensure that their network design is efficient, secure, and scalable.

Recap

- Network design is the process of planning and creating a network infrastructure tailored to meet an organization's specific needs, ensuring efficient and secure data flow.
- A well-designed network improves efficiency, scalability, security, cost-effectiveness, and user experience.
- Skipping network design planning leads to inefficiencies, security vulnerabilities, higher costs, and limited scalability.
- Essential steps and considerations to design a basic network structure include gathering requirements, assessing the current state, defining network goals, selecting devices and technologies, designing the physical network, creating an implementation plan, and documenting the network design.
- IBM's frequently used network design tools include Netcool Network Management, Cloud Pak for Network Automation, Spectrum Control, and SevOne Network Performance Management.
- Other vendor network design tools include Cisco Network Assistant, SolarWinds Network Topology Mapper, NetBrain, Visio, GNS3, and Wireshark (for real-time network traffic analysis).

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